Exploratory Data Analysis Using Python

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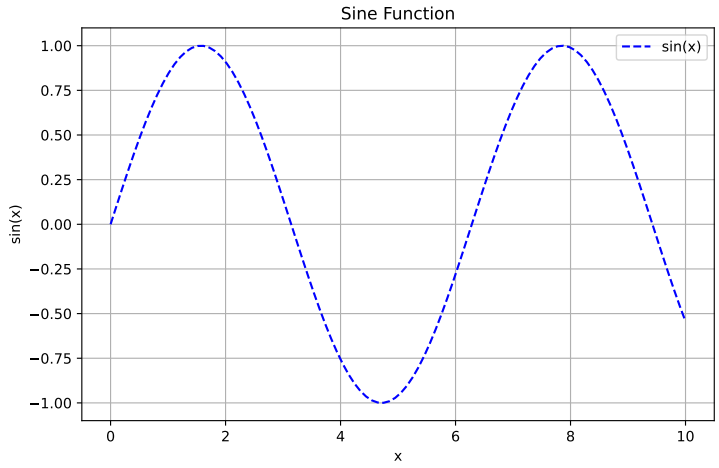


Figure 1: Random caption

1 Introduction

Working with some plots for demonstrational purposes.

1.1 Objectives

Made some graphs using matplotlib and numpy.

Graphs Graphs are genuinely great like 1.

2 Text Formatting

Types of texts **bold**, *italic*, underlined, and `monospaced` text.
More types of texts SMALL CAPITALS and *emphasized text*.

2.1 Font Sizes

Tiny text
Small text
Normal text
Large text
Huge text

2.2 Font Families

This is the default Roman font.
This is sans-serif text.
`This is typewriter text.`
This is bold and italic.

3 Types of charts

3.1 One

- Line Plot
- Bar Chart
- Histogram
- Scatter Plot

3.2 Two

1. Line Plot
2. Bar Chart
3. Histogram
4. Scatter Plot

3.3 Three

Line Plot Shows trends over time.

Histogram Shows frequency distributions.

Scatter Plot Shows relationships between variables.

3.4 Four

- A
    1. Aa
    2. Ab
- B
    1. Ba
    2. Bb

4 Graphs

Python generated plots.

4.1 Line Plot

2 shows the sine function.

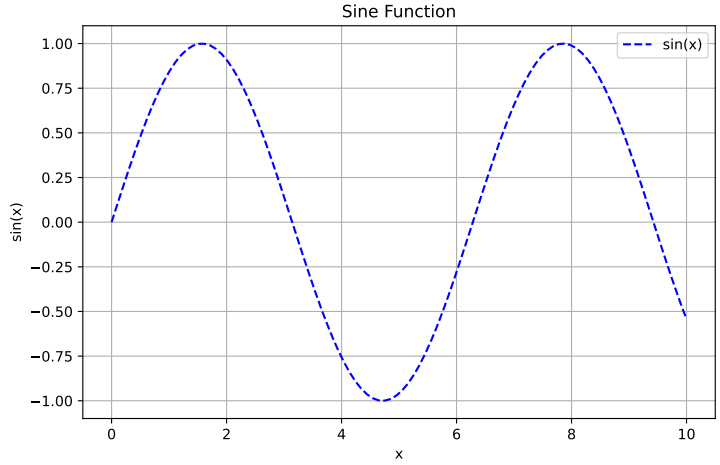


Figure 2: Line plot of the sine function

4.2 Bar Chart

3 Random bar chart.

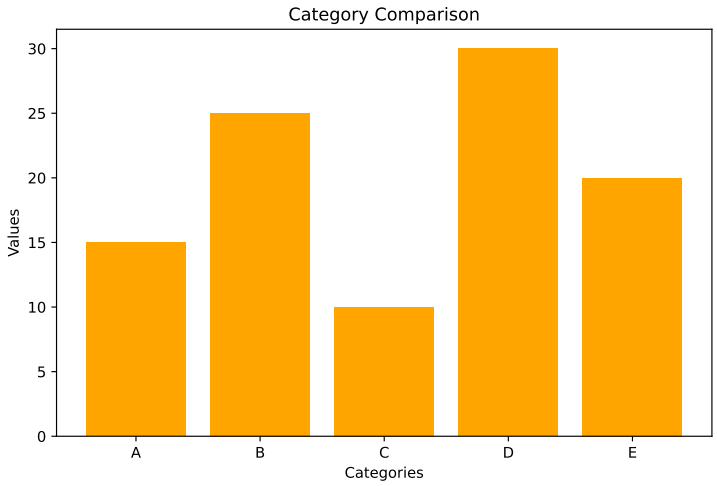


Figure 3: Random caption

4.3 Histogram

4 Random Histogram.

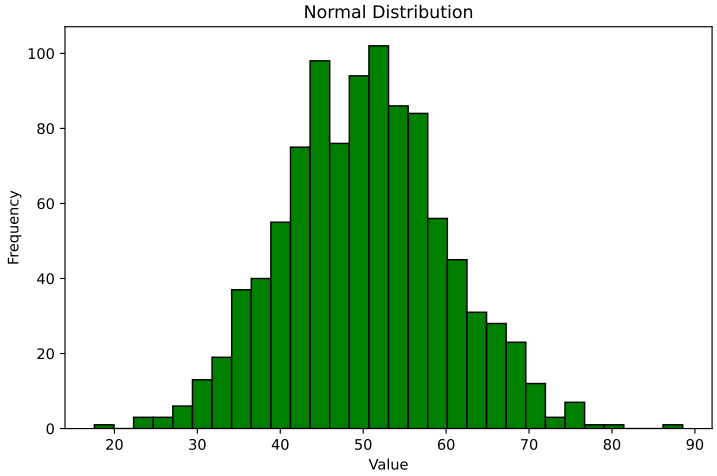


Figure 4: Random caption

4.4 Scatter Plot

5 Random scatter plot.

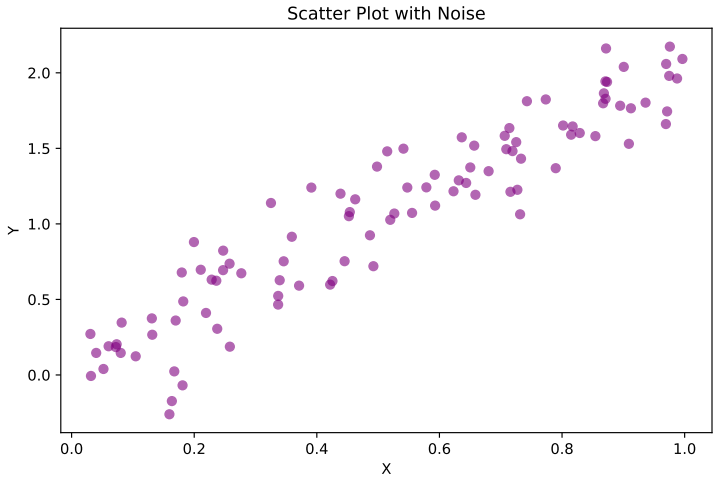


Figure 5: Random caption

5 Complex math stuff

5.1 Equations

Mean: $\mu = \frac{1}{n} \sum_{i=1}^{n} x_i$.
Mean but pretty.

$$\mu = \frac{1}{n} \sum_{i=1}^{n} x_i \tag{1}$$

5.2 Matrix

Behold matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \tag{2}$$

5.3 Multilined equations:

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x+1)^2 \end{aligned} \tag{3}$$

TANK U